## Abstract:

Asansol Municipal Cooperation is the considerable urban agglomeration centre of West Bengal in India. Rapid urbanization of the AMC has an impact on urban landscape and accelerates the high rise of surface temperature throughout the years. This study attempts to evaluate the trend of post monsoonal surface temperature of AMC over the periods by scrutinizing the changing scenario of vegetation cover and built up areas with the help of NDVI, NDWI and NDBI. LANDSAT 4-5 TM & LANDSAT 8 OLI/TIRS data have been used in this study to calculate the relevant indices. Maps and diagrams are made by using GIS Software and excel software. By analyzing all relevant data, maps and diagrams, it is clearly observed that due to dynamic urbanization, growing industrialization and cutting of trees rigorously the amount of dense vegetation cover has decreased and the built up area has been increased from core to periphery region. According to Shannon's entropy, the maximum value (1990) was recorded in 23 wards, however throughout a period of thirty years; this entropy value has been changed dramatically over 50 wards of AMC. So, there is a strong relationship between urbanization and LST, as a result the industrial area and core of city being a heat absorber has a maximum surface temperature of 30℃ and comparatively the fringe area has a range of 19-24℃ surface temperature. Hence, mean temperature has increased almost 3 to 5℃ within the study period. Pearson's correlation analysis for the measured parameters is presented high and poor correlation groups in the study area. A brief description of the methods and calculation of LST, NDVI, NDWI, NDBI and correlation are mentioned. The data can be reused for identifying the major problems associated with land surface temperature, reducing the urban green space and it is also helpful to the authorities to make policies to mitigate the problem associated with land surface temperature.